Adagio Institute, Inc.
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First-Round Questions to the Candidate

Paper 2

Koopman Spectral Theory for Indivisible Stochastic Processes

Candidate:	Benjamin D. Summers
Degree:	Doctor of Philosophy in Mathematical Physics (Concentration in Stochastic Quantum Theory), by Published Works
Doctoral Examiner:	E. F. Boero Adagio Institute, Inc.
Independent evaluation:	Dr S. R. Ismail-Sutton, Durham University
Identifiers:	Q-P02-01 to Q-P02-09 (technical report); Q-PF-05, Q-PF-09 (portfolio-level evaluation)
Compiled:	14 October 2026

This document belongs to the examination record (03_examiner_questions). It lists the questions put to the candidate in the first round of the examination of Paper 2 and gives each a stable identifier, by which the Report, the Technical Annex and the correspondence can cite it.

1 Sources and identifiers

Technical report. Section 2 transcribes Section 8, "Questions to the author", of the Doctoral Examiner's technical report on Paper 2, `TR-P02-DOCv2-EX01-Rv1.pdf`. The report identifies the version examined as SSRN, DOI 10.2139/ssrn.6733181, 3 September 2026, and carries the evaluation date 4 September 2026. The Doctoral Examiner sent it to the candidate on 7 September 2026 (COR-P02-03), and the candidate answered, to both examiners, on 24 September 2026 (COR-P02-05).

Identifiers. `Q-P02-`k is Question k of that report. The candidate's letters and the Technical Annex cite the questions by these numbers. The text is the report's, word for word; its section signs (§) refer to sections of the paper, and its finding numbers to findings of the report. Notes in small type are not part of the questions. They identify unresolved references, later corrections and related questions.

Independent evaluation. The independent evaluator's review of Paper 2 (`TR-P02-DOCv1-EX02-Rv1.pdf`, Exhibit J, 02_technical_evaluations) states its points as comments and poses no questions. It was sent to the candidate on 2 September 2026 and is filed as COR-P02-01; the candidate answered it on 3 September 2026 (COR-P02-02). It is not transcribed here.

Portfolio-level evaluation. Section 3 transcribes the questions of the portfolio-level technical evaluation (version 2, 4 September 2026; Exhibit H, Section 7, 02_technical_evaluations) that

concern Paper 2. `Q-PF-`k is Question k of that evaluation and has the same identifier in all six documents; Q-PF-09 concerns all six papers and appears in each. The evaluation names the works W1–W6, and Wn is Paper n.

2 Questions of the technical report

Source: `TR-P02-DOCv2-EX01-Rv1.pdf`, Section 8, Questions 1–9.

Q-P02-01 Explain precisely what is claimed to be original about $\beta(s) = C(s) - L(s)^2 - L'(s)$, given that $U'' = (A' + A^2)U$ is standard for differentiable non-autonomous evolution families, and attribute the expansion to Magnus/Dyson. State explicitly that the new content is the residual, the triple-scaling factor $(t-r)(r-s)$, and the integration into the deficit theory.

Q-P02-02 Confirm or refute Finding 4 — that the constant 1/3 in Corollary 4.6 is optimal, with extremal family $\Gamma(t,r) = \Gamma(r,s) = I$, $\Gamma(t,s) = I - \frac{D}{2}(I - X)$, the infimum being approached as $D \to 0$ — and if confirmed, incorporate it, closing Remark 4.7. State also that the bound is attained only at a degenerate corner of the class, so that its practical looseness on smooth families is not misread.

Note. Question 2 says that the bound is attained at a degenerate corner of the class. The candidate's Proposition 4.7 shows that no family attains 1/3: the constant is optimal only as an infimum, approached as $D \to 0$. The Doctoral Examiner corrected his formulation in his letter of 25 September 2026 (COR-P02-06; Technical Annex, Section T.2(e)).

Q-P02-03 State the strongest priority claim you would defend for V_{in} and V_{out}, and identify the closest operator-theoretic notion of a defect space that you consider prior art.

Q-P02-04 Address Doukas (arXiv:2602.22095) and Canturk (arXiv:2503.09333; *Ann. Phys.* **487**, 170367), both predating the deposit, and state what remains your own in §5 once the basic stochastic-matrix-to-CPTP lift is conceded to that literature.

Q-P02-05 Clarify whether any feature of the formal Hamiltonian is invariant under changes of Stinespring dilation or of unitary completion. If not, which quantities derived from the lift are intended to carry physical status?

Q-P02-06 Explain the exact relationship between your finite-grid cocycle dilation and standard process-tensor or repeated-interaction/collision-model dilations.

Q-P02-07 Give a concrete statistical protocol by which Δ or β could be estimated from finite transition data, and state what null hypothesis would be tested. In doing so, address the conditioning problem recorded in the verification record above: β is a small difference of large second-order quantities, so any estimator will lose several significant digits to cancellation. State what accuracy in Γ would be required to resolve β at a given size.

Note. The report in the form supplied contains no verification record, although its Section 2 announces one ("the verification record is set out below"), and the conditioning problem

appears in it only in this question. The portfolio-level evaluation (Exhibit H, Section 2.4(i)) records the problem: in a finite-difference check of Proposition 4.12, $\|\beta\| \approx 0.70$ arose as a difference of quantities of size about 10, so more than one significant digit was lost to cancellation.

Q-P02-08 Describe what an empirically measured CK violation can and cannot establish about intrinsic memory once hidden-state augmentation and coarse-graining are allowed.

Q-P02-09 Exhibit the framework on at least one family of dimension greater than three, or state why the two- and three-state cases are structurally sufficient.

3 Questions of the portfolio-level evaluation

Source: portfolio-level technical evaluation, version 2, 4 September 2026 (Exhibit H, exhibits/H-technical-evaluation), Section 7. Question 5 concerns Paper 2; Question 9 concerns all six papers.

Q-PF-05 **W2 Remark 4.7.** The examination finds the constant $1/3$ in Corollary 4.6 optimal, with extremal family $\Gamma(t, r) = \Gamma(r, s) = I$, $\Gamma(t, s) = I - (D/2)(I - X)$, the infimum approached as $D \to 0$. Confirm or refute, and incorporate.

Note. Q-P02-02 asks the same question. The note to Q-P02-02 applies: the constant is optimal only as an infimum and no family attains it.

Q-PF-09 **Deposit venue.** State whether the works will be deposited on arXiv (math-ph / math.PR / quant-ph).